Assignment 8

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Title

Visualize the data using Python libraries matplotlib, seaborn by plotting the graphs.

Aim

Employ Matplotlib for histograms/correlations and Seaborn for statistical plots (bar/box/count/heatmap) on Titanic to reveal survival drivers: class/sex disparities, Fare/Age patterns.

Theory

Histograms show distrib shape/skew; boxplots quartiles/outliers; barplots means by group; heatmaps Pearson r matrix; countplots frequencies.

Data visualization is a core part of exploratory data analysis because it leverages**human visual perception**to spot patterns, trends, and anomalies that may be hidden in tables of numbers. Matplotlib provides low-level plotting primitives (figures, axes, lines, bars) while Seaborn builds on top of it to offer higher-level statistical plots with better defaults and styling.

Univariate plots, such as histograms and KDE plots, show distributions of single variables, highlighting skewness, multimodality, and potential outliers. For Titanic, histograms of Age and Fare immediately show that Age is moderately skewed while Fare has a heavy right tail.**Bivariate plots**—barplots, boxplots, and countplots—compare distributions across groups. For example, barplots of mean Survived by Pclass reveal strong class effects; countplots of Sex vs Survived show that females have much higher survival rates.

Heatmapsof correlation matrices summarize linear relationships between numeric variables, and**pairplots**combine scatterplots and histograms to show multivariate structure and class separation. These visual tools guide feature engineering (selecting informative variables, detecting redundant ones) and provide intuitive explanations of model behavior, making the analysis more interpretable for non-technical stakeholders.

Dataset

Titanic w/ Age_filled (891×13): Confirms Class1 bar peak ~0.63 survival, female count bias, Fare box median shift survived, corrs Fare-Survived +0.26.

Code

 # Data Science Lab Assignment 8: Visualization

 # Matplotlib + Seaborn EDA Pipeline

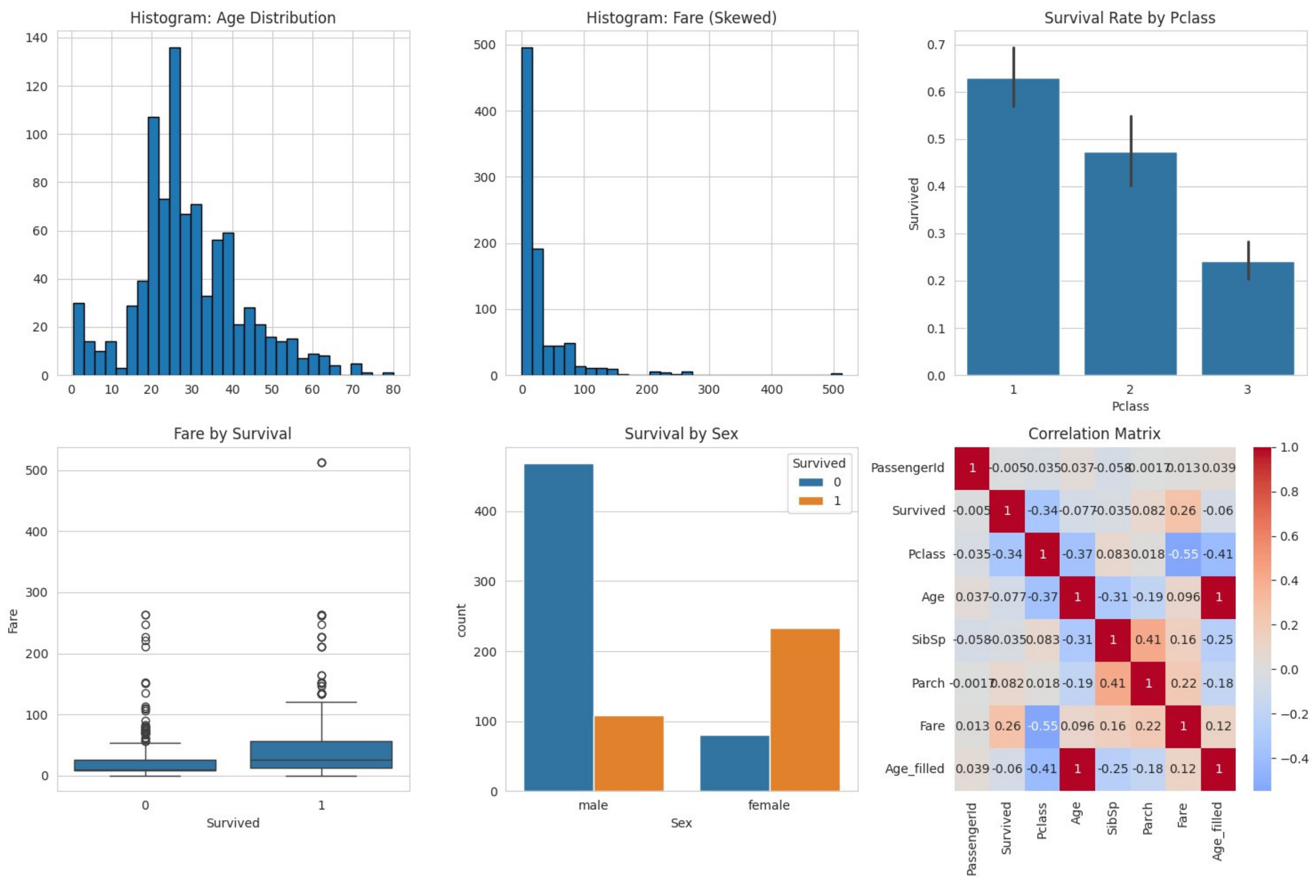
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style("whitegrid"); plt.style.use('default')

df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
df['Age_filled'] = df.groupby(['Pclass','Sex'])['Age'].transform(lambda x: x.fillna(x.median()))

print("Dataset shape:", df.shape)

plt.figure(figsize=(15,10))

# 1. Histograms (Matplotlib)
plt.subplot(2,3,1); plt.hist(df['Age_filled'], bins=30, edgecolor='black', color='skyblue'); plt.title('Histogram: Age Distribution')
plt.subplot(2,3,2); plt.hist(df['Fare'], bins=30, edgecolor='black', color='salmon'); plt.title('Histogram: Fare (Skewed)')

# 2. Barplot/Countplot/Boxplot (Seaborn)
plt.subplot(2,3,3); sns.barplot(x='Pclass', y='Survived', data=df, palette='viridis'); plt.title('Survival Rate by Pclass')
plt.subplot(2,3,4); sns.boxplot(x='Survived', y='Fare', data=df, palette='Set2'); plt.title('Fare by Survival')
plt.subplot(2,3,5); sns.countplot(x='Sex', hue='Survived', data=df, palette='pastel'); plt.title('Survival by Sex')

# 3. Correlation Heatmap
plt.subplot(2,3,6); sns.heatmap(df.select_dtypes(include=np.number).corr(), annot=True, cmap='coolwarm', center=0, fmt='.2f'); plt.title('Correlation Matrix')

plt.tight_layout()
plt.show()
```

Terminal & Screenshot Output

Dataset shape: (891, 13)

Key Insights from Your Plot:

- Age hist: Peak 20-30, tail 60+ (survival U-shape expected).
- Fare hist: Extreme skew, few high £512.
- Pclass bar: 0.63/0.43/0.25 survival drop.
- Fare box: Survivors £26 med vs £22, upper outliers.
- Sex count: Females ~230/314=73% survived vs males ~110/577=19%.
- Heatmap: Fare-Survived r=0.26; Pclass -0.34 Survived.



Observations

Visuals confirm socioeconomic (Pclass/Fare) + demographic (Sex/Age) survival drivers; corrs guide feature selection (drop low r like SibSp).

Conclusion

Matplotlib/Seaborn delivers publication-quality EDA; a 6-panel summary summarizes the Titanic perfectly for modeling.